

Name: White, Rice, & Spice Team

Date: 3/10/26

Class: Design for Engineers e4usa

MyDesign Portfolio

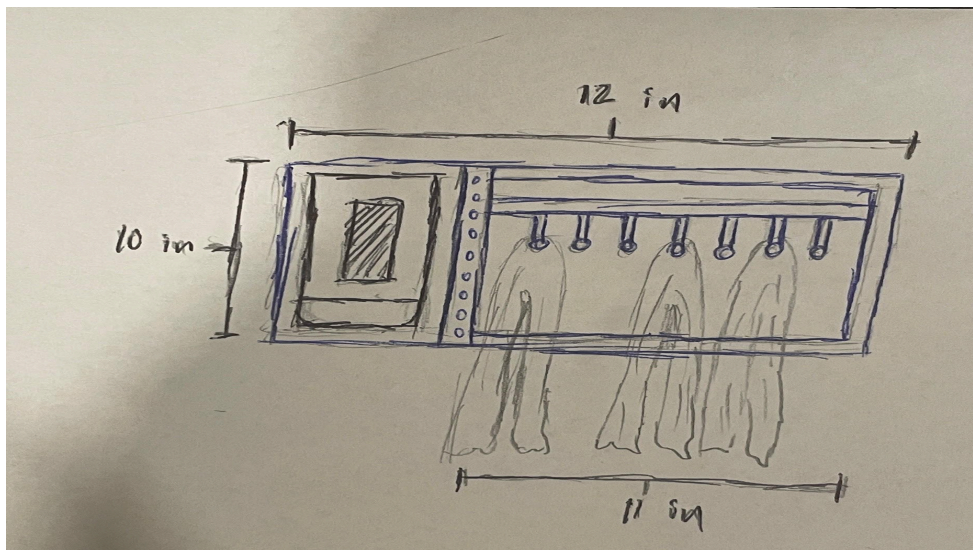
Element D: Design Concept Generation, Analysis, and Selection

Design Idea Generation (D1)

Towel Drying Rack

Date: 3/3/26

Description: The design is a rack fixed to the wall for towels to be hung to dry. After being used to clean up, the residual soap on the towels will be contained in the rack to dry and be reused.



Soap Controller

Date: 3/3/26

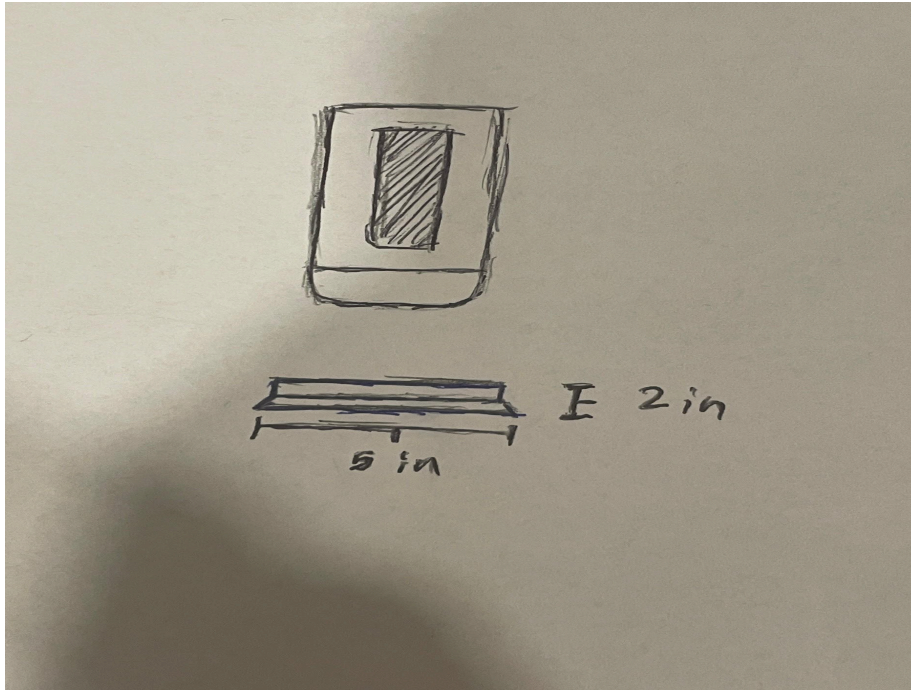
Description: This design creates a nozzle or tube over the soap dispenser to contract the

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flow of soap, ultimately using less spilled soap to prevent puddles from forming.



Air Drying System

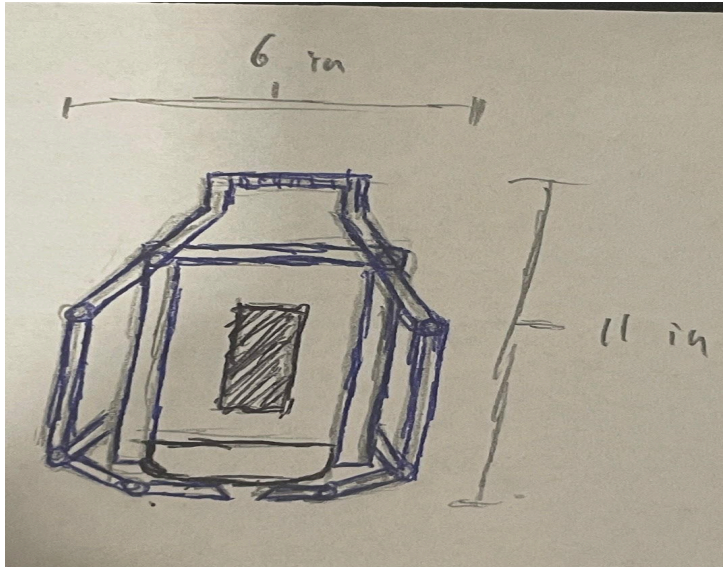
Date: 3/3/26

Description: The design involves letting the soap dry & evaporate on a predetermined surface.

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3D Printed Structure:

Date: 3/3/26

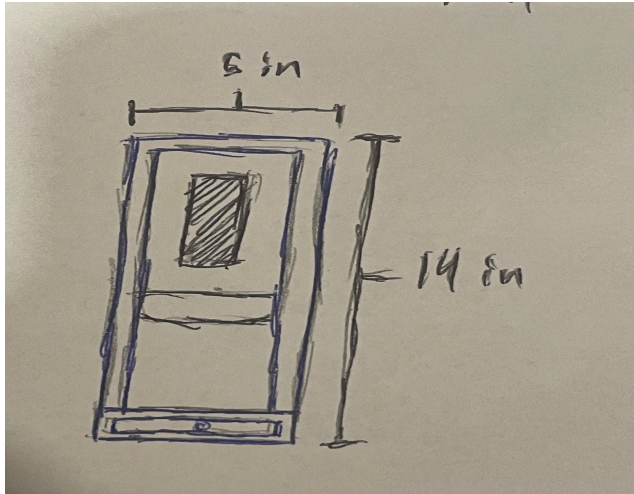
Description: The structure includes multiple features, including a funnel to collect soap, a container, a hook or attachable apparatus, & a mechanism to easily dispose of the excess, all using 3D printing construction methods.

Drawing that includes distinguishing information such as scale, dimensions, and materials:

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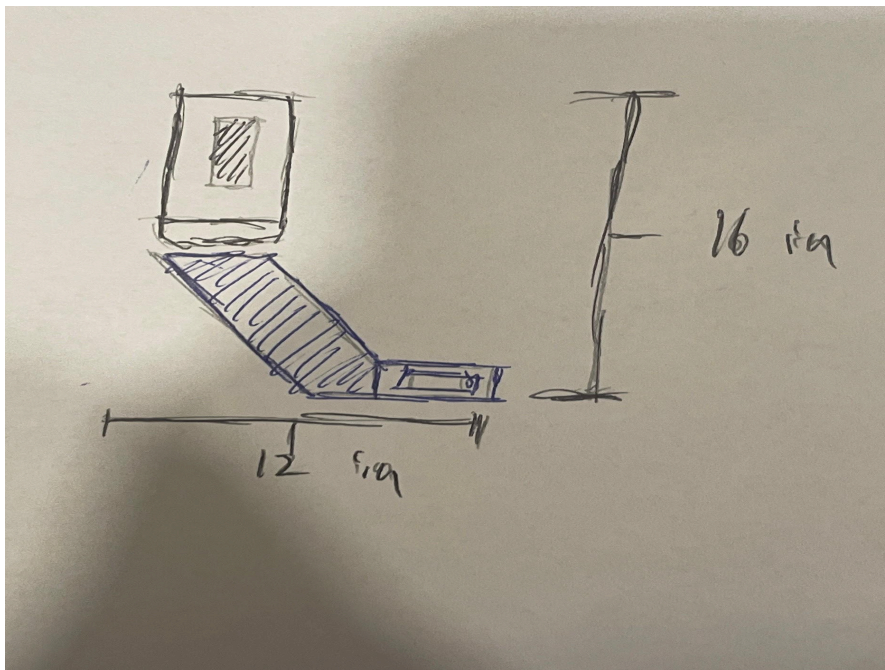
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Ramp Structure

Date: 3/3/26

Description: A ramp structure will guide the excess soap into a predetermined area or container.



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Design Idea Comparison and Selection (D1, D2)

	Weight	Towel drying rack	Soap controller	Air drying	3D printed structure	Ramp structure
Quality materials	10	5	10	5	10	10
Ease of application	10	0	5	0	10	10
Puddle prevention	30	15	20	15	30	15
Proper measurements	25	12.5	20	12.5	25	25
Supports weight	25	15	25	12.5	25	12.5
Other (is this design possible/reasonable)?	100	50	100	50	100	100
score	200	97.5	180	95	200	172.5

The team chooses the 3D Printed Structure to solve the community problem.

Design Blueprint (D4)

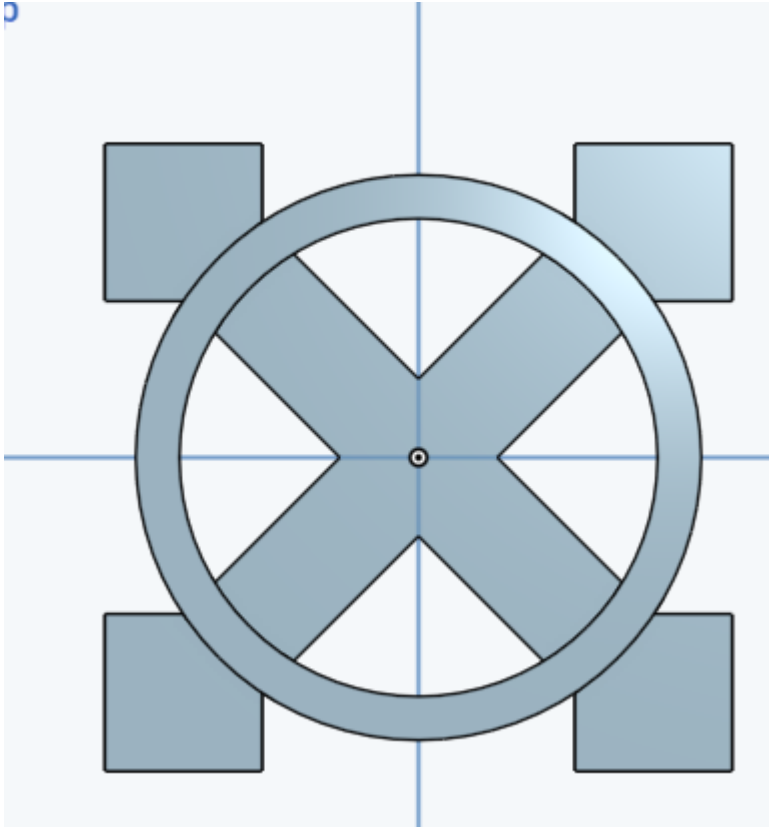
Design Blueprint #1

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Date: 3/5/26

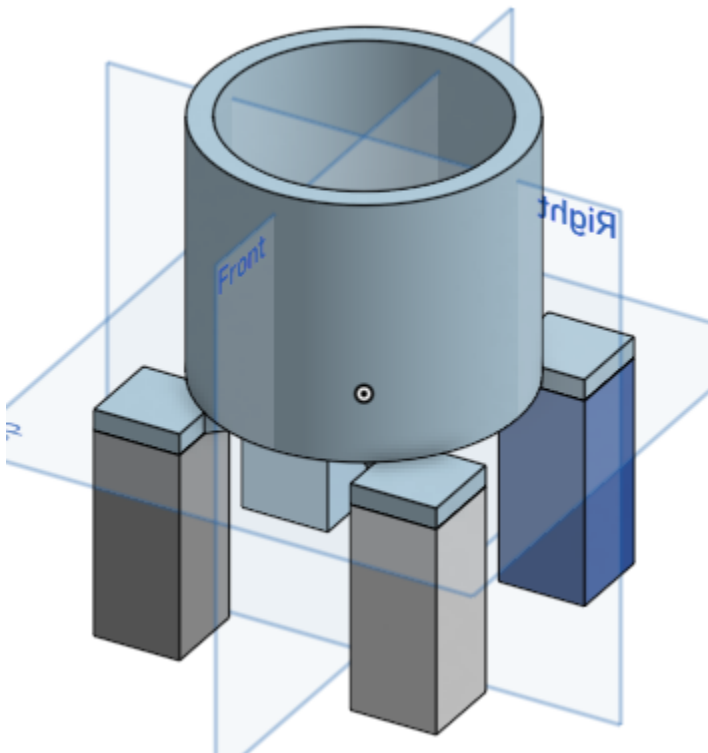
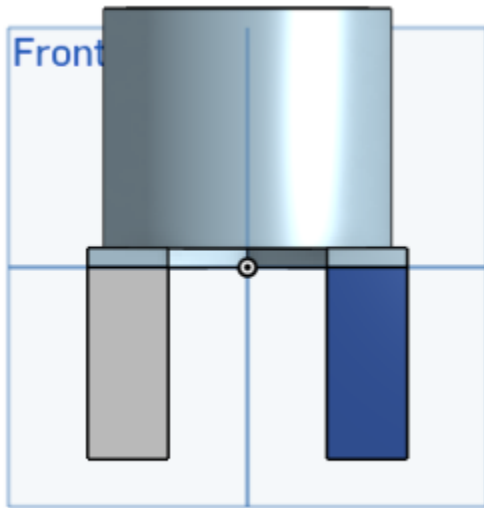


Sketch:

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Manufacturing Methods:

MyDesign Portfolio - Element D Initial Template

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Step 1: 3D printing the materials/parts

Step 2: Assembling the parts

Step 3: Attach the design to the soap dispenser

Supply List and Costs: (Not required for programming)

Item	Cost each	# needed	Total
Base of Design, 3D printed	N/A, school-provided materials and 3D printers.	1	\$0
Pillar of Design, 3D printed	N/A, school-provided materials and 3D printers	2	\$0
Soap holder Design, 3D printed	N/A, school-provided materials and 3D printers	1	\$0
Soap dispenser	\$2.99 from IKEA	1	\$2.99
Total Cost	4 parts	4	\$2.99

Plan of Action (D3)

White, Rice, & Spice Team Ghant Chart:



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1	Project Conception and Initiation					
3.2.3	Team Charter					100%
3.3.3	Team Charter Revisions					100%
3.5.1	Research - for Element A and B					100%
3.4.1	Stakeholders - Who communicates with them?					100%
	Element A					100%
	Element B					100%
	Element C					100%
2	Project Planning					
3.6	Brainstorming and Selecting Ideas					100%
3.7.1	Engineering Drawings					100%
3.8.3	Communication Plan - Google, Group chat, Notebook					100%
	Risk Management - What could go wrong with the Prototype					100%
	Element D					100%
	Element E					100%
	Element F					100%
3	Project Conception and Initiation					
3.8.1	Prototype Creation					100%
3.8.2	Prototype Testing					100%
	Prototype Results					100%
	Element G					100%
	Element H					100%
	Element I					100%
4	Project Performance / Monitoring					
3.10.1	Poster Planning					100%
	Poster Creation					100%
	Poster Editing					100%
	Project Performance					100%
5	Project Reflections					
4.1	Feedback - Element J					100%
4.2.1	Project Planning Revisit					100%
	Project Iteration					100%
4.3.1	Project Iteration Presentation					100%
	Element H					100%
	Element I					100%

